MIL-B-43909(GL)

26 December 1974
SUPERSEDING
(See 6.3)

MILITARY SPECIFICATION

BELLOWS, EXPANDABLE SHELTER (MUST)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an insulated bellows for the MUST complex, fabricated from coated cloth and polyurethane foam.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester
DDD-L-20	-	Label, for Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-H-639	-	Hooks (for Belts, Straps, and Equipage Items)
MIL-R-3390	-	Rings, Dee
MIL-W-4088	-	Webbing, Textile, Woven Nylon
MIL-G-16491	-	Grommets, Metallic
MIL-C-43285	-	Cloth, Coated (Chloroprene Base Coated, Chlorosulphonated Polyethylene Top Coated)

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods
FED-STD-595	-	Colors
FED-STD-751	-	Stitches, Seams, and Stitchings

FSC 5410

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MS-63033 - Eyelet, Metallic, Rolled Deep Flange, with Washer, Style A

DRAWING

ARMY NATICK LABORATORIES

5-4-1650 - Bellows, Expandable Shelter (MUST); Assembly Complete

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials (ASTM)

D 1564 - Testing Slab Flexible Urethane Foam

(Application for copies should be addressed to the American Society for testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Freight Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Alternate components. Components offered as equivalent to components specified hereinafter and on the reference drawing as a specific manufacturer's part number or equal shall be functionally equal, of equal or better quality, and of equal or less weight with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article, the supplier shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified above. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article manufacture and does not relieve the supplier of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), the supplier shall furnish a sample of the complete bellows for first article inspection and approval (see 4.2 and 6.4).

3.3 Materials and components. Materials and components shall conform to the specifications, standards and drawing listed in section 2 and as specified herein.

3.3.1 Polyurethane foam. The polyurethane foam shall be a flexible and flame-retardant foam conforming to the following requirements when tested as specified in 4.3.1.1.1.

Density, lb/cu ft	2.6 maximum
Indentation load, lb, deflection at 25 percent	25 ± 5
50 percent compression set	20 percent maximum
Tear resistance, lb - force per lineal inch	2 minimum
After-flame time, seconds	
Original	2 maximum
After oven aging	2 maximum
After steam autoclave	2 maximum

3.4 Construction. The bellows shall conform to the drawing listed in section 2 and as specified herein.

3.4.1 Stitching, machine. All stitching shall conform to type 301 of FED-STD-751, 5 to 7 stitches per inch.

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3.4.1.1 Thread breaks. Thread breaks shall be overstitched not less than 1 inch at each break.

3.4.1.2 Stitching ends. Ends of stitching shall be overstitched not less than 1 inch except where ends are turned under in a hem, held down by other stitching, or where stitching is performed automatically. Where stitching is performed automatically on stitch patterns such as box, box with cross stitch, "W" stitching or straight line tacking, three or more tying, overlapping or backstitches shall be used to secure the ends.

3.4.1.3 Skipped stitches. Two or more consecutively skipped stitches shall be overstitched not less than 1 inch.

3.4.1.4 Lubricant. The addition of lubricant to the polyester thread prior to or during the sewing operations is prohibited.

3.4.2 Setting of grommets. Holes punched to receive grommets shall be smaller then the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommets shall be securely clinched without cutting the materials.

3.4.3 Fusing of webbing ends. Cut ends of nylon webbing shall be fused and, after fusing, shall be free from frayed edges and loose fibers.

3.5 Marking. The application of the letters "US" and the nomenclature on the bellows shall conform to type IV, class 9 of DDD-L-20 in the size characters and in the locations shown on the drawing. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

3.5.1 Identification labels. Identification labels shall conform to type II, class 6 of DDD-L-20 and shall be 3-1/2 inches by 5-1/2 inches minimum with a blank space of at least 1-1/4 inches provided on the bottom of the label for the Department of Defense Inspector's stamp. Identification labels shall be stitched to the bellows in the location shown on the drawing.

3.6 Workmanship. Cloth components shall be clean and free of holes, cuts, tears, scratch or abrasion marks or cloth defects such as uncoated areas, blisters, tunnels, foreign matter or delamination of coating, lumps or heavily coated areas. Cord components shall be free of cuts or abrasions. Webbing and tapes shall not have frayed or scalloped edges. Thread tension shall be maintained so that there will be no loose or tight stitching. Seam allowances shall be maintained with seams properly sewed so that no run-offs, pleats or open seams shall result. Care shall be taken to see that no needle chews occur. All thread ends shall be trimmed to 1/2 inch or less. Metal components shall be free of burrs, sharp edges, corroded areas and shall not be broken or malformed.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. When a first article is required, it shall be examined for defects listed in tables I and II. The presence of any visual defect or any dimension not within the specified requirements shall be cause for rejection of the first article.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and materials. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Material test.

4.3.1.1.1 Polyurethane foam. The polyurethane foam shall be tested as follows for compliance with the requirements of 3.3.1. Any nonconformance shall be cause for rejection of the lot of material.

Characteristic	Test method
Density	ASTM D 1564, sec 69-74
Indentation load, deflection	ASTM D 1564, sec 19-25
50 percent compression set	ASTM D 1564, sec 12-18
Tear resistance	ASTM D 1564, sec 75-81
After flame time	Method 5905 FED-STD-191
Original	ASTM D 1564, sec 45-49.2
After oven aging, 22 hours at 284°F	ASTM D 1564, sec 5-9.5
After steam autoclave, 5 hours at 250°F	

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4.3.1.1.2 Lot. For purposes of sampling, a lot shall consist of all material produced under like conditions in one unchanged process by one manufacturer and submitted for acceptance at one time.

4.3.1.1.3 Sampling for testing. Sample shall be selected as listed below and subjected to the tests described in 4.3.1.1.1

a. Rolls - A sample roll shall be selected at random from the lot. Specimens of sufficient size to conduct the tests shall be taken from the sample after removal of two thicknesses of material from the roll.

b. Sheets - Three bundles shall be selected, the top three sheets of each bundle removed, and sample taken at random from the remainder of each bundle.

4.3.2 In-process inspection. Inspection shall be made to determine that the following are as specified. Whenever nonconformance is noted, correction shall be made to the items affected and the process.

a. There shall be no lubrication of polyester thread prior to or during sewing operations for compliance with 3.4.1.4.

b. Holes punched to receive grommets are in compliance with 3.4.2.

4.3.3 End item inspection. The lot shall consist of all bellows offered for inspection at one time. The sample unit shall be one bellows.

4.3.3.1 Visual examination. Visual examination of the bellows shall be made for defects in table I. The inspection level shall be II and the acceptable quality levels (AQL's), expressed in terms of defects per hundred units, shall be 15.0 major, 65.0 major and minor A defects combined and 100.0 total major, minor A and B defects combined.

TABLE I. Classification of defects for visual characteristics

Examine	Defect	Classification Major	Minor A	Minor B
Fabric	Any hole, cut, tear	X		
	Exposed fabric, delamination of coating	X		
	Scratch or abrasion, blister, tunnel, crease or wrinkle		X	
	Lump or heavy coated area			X
Webbing	Edge frayed or scalloped; cut end not fused			X
Hardware	Broken	X		

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Hardware (cont'd)	Malformed:			
	- fails to perform intended function	X		
	- but will perform intended function		X	
	Corroded area			X
	Burr or sharp edge which may cause injury in handling or damage to fabric	X		
Grommets	Improperly set, two or more teeth exposed			X
	Clinched loosely, allowing grommet to rotate around hole			X
	Clinched excessively tight; i.e., cutting fabric	X		
	Insecurely clinched; i.e., grommets or washers become disengaged	X		
Seams and stitching	Open seam:			
	- 1/2 inch or less			X
	- more than 1/2 inch, but not more than 1 inch		X	
	- more than 1 inch	X		
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken, or when one or more skipped or runoff stitches occur. On double stitched seams, a seam shall be classified as open when either one or both sides of a seam are open.			
	Thread breaks overstitched less than 1 inch			X
	Needle chews resulting in hole, cut or tear	X		
	Skipped stitches overstitched less than 1 inch (where 2 or more consecutively skipped stitches occur)			X
NOTE:	Skipped stitches not overstitched shall be classified as open seams.			
Seam type	Wrong seam type	X		
Stitch type	Wrong stitch type	X		
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches in length			X

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching (cont'd)				
Stitch tension (cont'd)	Tight, as evidenced by puckering on seams for more than 6 inches in length			X
Stitches per inch	Less than minimum specified:			X
	- one stitch		X	
	- two or more stitches			
	More than maximum specified:			X
	- one stitch		X	
	- two or more stitches			
	NOTE: Variation in the number of stitches caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or in turning corners, shall be classified as follows:			
	a. Within the minor "B" defect classification - no defect			
	b. Within the minor "A" defect classification - minor "B" defect			
Stitching margin	Larger than specified by more than 1/16 inch:			
	- for more than 1 inch, but not more than 4 inches in length			X
	- for more than 4 inches in length		X	
Stitching gage	Not as specified		X	
Stitching ends	Overstitched less than 1 inch (except where ends are held down by other stitching, turned under in a hem or where stitching is performed automatically)			X
	Secured with less than three tying, overlapping or backstitches when automatic stitching is performed			X

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TABLE I. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Components and assembly	Any component part not fabricated of applicable referenced material	X		
Marking	Omitted, incorrect, illegible, misplaced or size characters not as specified			X
Cleanness	Grease or oil stains			X
	Thread ends not trimmed to 1/2 inch or less			X

4.3.3.2 Dimensional examination. Dimensional examination of the bellows shall be made for defects in table II. The inspection level shall be S-3 and the AQL's, expressed in terms of defects per hundred units, shall be 2.5 major and 25.0 total major and minor defects combined.

TABLE II. Classification of defects for dimensional characteristics

Examine	Defect	Classification Major	Minor
Frame-flap assemblies	Assembled flap sections and corner pieces less than 262 inches or more than 266 inches in length	X	
Location dimensions	Not within specified tolerance		X

4.3.4 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements are in compliance with section 5. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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TABLE III. Examination of preparation for delivery

Examine	Defect
Marking	Omitted, incorrect, illegible, of improper size, location, sequence or method of application
Materials	Any component missing or damaged
Workmanship	Bulging or distortion of containers
Content	Number per container is more or less than required

4.3.5 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and reinforcing shall be in accordance with that specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each bellows shall be compactly folded and packaged in a snug-fitting fiberboard box conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 200, of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.1.2 Level C. Bellows shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Three bellows, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to grade V2s, style RSC of PPP-B-636. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B. Three bellows, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275, of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

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5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C. Bellows, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The bellows covered by this specification is intended for use as an interconnecting corridor or passageway between two MUST expandable shelters.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) First article (see 3.2, 4.2 and 6.4)
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1)

6.3 Supersession data. This specification includes the requirements of US Army Natick Laboratories (GL) Limited Purchase Description LP/P.DES 52-69 dated 22 December 1969.

6.4 First article. When a first article is required, it shall be inspected and approved under appropriate provisions of ASPR 7-104.55. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

Custodian:

Army - GL

Review activity:

Army - MD

Preparing activity:

Army - GL

Project No. 5410-A115

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